

RAM RACING

Aerodynamic Subteam

Wheel MRF Setup Guide

How to correctly create Moving Reference Frame zones in Ansys Discovery
for use with the Ram Racing CFD Automation Tool

This guide explains what Wheel MRF zones are, why we use them, and exactly how to create them in Ansys Discovery before exporting your .pmdb geometry file. Following these steps correctly is required for the automation tool to assign Moving Reference Frames to each wheel.

1. What is a Wheel MRF Zone?

MRF stands for **Moving Reference Frame**. Instead of spinning the wheel surface as a rotating wall boundary condition, MRF creates a **rotating fluid volume** around each wheel. The mesh inside that volume is treated as if it is rotating at the wheel's angular velocity.

Why MRF instead of rotating wall?

- Rotating wall only spins the surface — the air around the tire spins wrong.
- MRF rotates the entire flow field around the wheel — physically correct.
- Better captures the rotating wake behind open-wheel tires.
- Required for accurate front wing downforce predictions near tires.

The automation tool automatically calculates wheel RPM from vehicle speed and wheel radius using: **$RPM = (V / r) / (2\pi) \times 60$** where V is in m/s and r is wheel radius in meters.

2. Required Zone Names

The automation tool looks for specific zone names in Fluent. You **must** use these exact names in Discovery Named Selections, or update them in the tool's Zone Names tab.

Wheel	Default Zone Name	Rotation Axis Direction	Notes
Front Left	mrf_flw	+Z (axis_z = +1.0)	Driver's left side
Front Right	mrf_frw	-Z (axis_z = -1.0)	Driver's right side
Rear Left	mrf_rlw	+Z (axis_z = +1.0)	
Rear Right	mrf_rrw	-Z (axis_z = -1.0)	
Half Car (Right only)	mrf_frw + mrf_rrw	+Z for both	Symmetry at z=0

For a **half-car simulation** you only need two zones: `mrf_frw` (front right) and `mrf_rrw` (rear right). The symmetry plane is at $z = 0$.

3. Creating MRF Zones in Ansys Discovery

These steps are done **after** all your geometry simplification is complete and **before** you export the .pmdb file. Each wheel needs its own MRF enclosure — a cylindrical or box-shaped fluid volume surrounding the wheel.

Step 1 — Fillet the Wheels

Before creating MRF zones, fillet all sharp wheel edges to approximately 0.5 inch radius. This prevents mesh failures at the wheel-ground contact patch and helps convergence. Use the Fillet tool in Discovery on all wheel edge loops.

Step 2 — Note Wheel Center Coordinates

Find the exact center coordinates of each wheel in meters. You will need X, Y, Z coordinates for each wheel axle. In Discovery, use the Measure tool or check your SolidWorks assembly. Write these down now — you will enter them in the automation tool's Wheel MRF tab.

Step 3 — Create a Cylinder Around the First Wheel

Go to: Create > Cylinder (or use the Enclosure tool). Create a cylinder centered on the wheel axle. The cylinder should: (a) be slightly larger than the wheel — extend ~20mm past the tire in all directions, (b) be oriented with its axis along the wheel axle (Z axis in Ram Racing convention), (c) NOT intersect the ground plane. A good rule of thumb: diameter = wheel_diameter + 40mm, height = wheel_width + 40mm.

Step 4 — Boolean Subtract to Create Fluid Volume

The MRF zone must be a fluid region inside your main enclosure. If you used the Enclosure workflow in Discovery, your cylinder is already inside the fluid enclosure. Ensure there is no solid material inside the MRF cylinder — it should be pure fluid surrounding the wheel exterior.

Step 5 — Create a Named Selection for the MRF Zone

Select the cylindrical volume (cell body). Right-click > Named Selection > New Named Selection. Name it exactly: `mrf_frw`, `mrf_flw`, `mrf_rrw`, or `mrf_rlw` (matching the table in Section 2). This name will be recognized by Fluent as a separate cell zone.

Step 6 — Repeat for Each Wheel

Repeat Steps 3-5 for every wheel in your simulation. For a half-car simulation you need two zones (`mrf_frw` and `mrf_rrw`). For a full-car simulation you need four zones. Make sure the cylinders do not overlap each other.

Step 7 — Verify the Enclosure Structure

Your geometry should now have: the main fluid enclosure (`enclosure-enclosure1`), MRF cylinder zones inside it (`mrf_frw`, etc.), and the car body inside the main enclosure. Use the Body Structure panel to verify

all bodies are present.

Step 8 — Export as PMDB

File > Export > Export as PMDB. Name the file descriptively (e.g. HalfCar_V3_MRF.pmdb). This is the file you will select in the automation tool's geometry field.

4. MRF Zone Sizing Reference

Use these guidelines for cylinder dimensions. Values are for a typical FSAE open-wheel car with ~8 inch radius tires.

Parameter	Formula	Typical Value	Notes
Cylinder Radius	wheel_radius + 20mm	~225mm	Must clear tire bulge
Cylinder Height	wheel_width + 40mm	~230mm	Centered on axle
Center X	from axle position	varies	Enter in tool Wheel MRF tab
Center Y	axle height from ground	~152mm	~6 inches for FSAE
Center Z	lateral axle offset	varies	+Z = left, -Z = right
Rotation Axis	wheel axle direction	(0, 0, ±1)	Z-axis; sign per side
RPM	auto: $V/r/2\pi*60$	~1000 RPM @ 40mph	Set 0 in tool for auto

5. What Happens During Meshing

When the automation tool runs meshing with your .pmdb file, it will:

- Detect each mrf_xxx named selection as a separate cell zone.
- Apply the wheel curvature sizing (0.5mm min, 32mm max, 18 deg angle) to the tire surfaces.
- Add a local refinement box around each wheel (32mm mesh size).
- NOT add boundary layers inside the MRF zones (only on aero devices and ground).

During solver setup, the tool will:

- Set each mrf_xxx cell zone to Moving Reference Frame type.
- Calculate RPM from vehicle speed and wheel radius automatically.
- Set the rotation center to the coordinates you enter in the Wheel MRF tab.
- Set rotation axis direction from your axis_z setting (+1 or -1).

6. Common Mistakes

✗ MRF cylinder intersects the ground

The cylinder bottom must NOT touch $z=0$. Leave at least 5mm gap. If it touches the ground, the mesh will fail at the intersection.

✗ Wrong zone name

Zone names are case-sensitive. 'MRF_FRW' will not be found — use 'mrf_frw'. Check in the automation tool's Zone Names tab that names match exactly.

X MRF zones overlap each other

Cylinders for different wheels must not intersect. Check this visually in Discovery before exporting.

X MRF zone too large

The cylinder should be just large enough to surround the tire. If it is too large it will overlap with other geometry or the ground plane, and the mesh quality will degrade near the boundaries.

X Wheel not filleted

Always fillet the wheel edges before creating MRF zones. Sharp edges on the tire cause bad surface mesh elements and convergence failure.

X Forgetting to set center coordinates

The rotation center must be at the wheel axle center in meters. If you leave it at 0,0,0 the wheel will spin about the origin, not the axle, and the simulation will diverge or produce garbage.

7. Entering Values in the Automation Tool

In the **Wheel MRF tab** of the simulation editor, for each wheel enter:

- **Wheel Name** — FLW, FRW, RLW, or RRW (just a label).
- **Fluent Zone Name** — exactly mrf_flw, mrf_frw, mrf_rlw, or mrf_rrw.
- **Center X/Y/Z** — wheel axle center position in meters.
- **Axis X/Y/Z** — rotation axis direction. For Z-axis: set Axis Z = +1 (left wheels) or -1 (right wheels). Leave Axis X and Axis Y at 0.
- **Wheel Radius [m]** — outer radius of the tire, e.g. 0.203m (~8 inch).
- **RPM** — set to 0 for automatic calculation from vehicle speed.

The auto RPM formula: $RPM = (V [m/s] / R [m]) / (2 \times 3.14159) \times 60$. At 40 mph (17.88 m/s) with $R=0.203m$, this gives approximately 840 RPM.